

DiffGerber

Decision-Focused Learning of the Gerber Threshold

DiffGerber learns the one hand-set threshold inside the Gerber risk model from realized portfolio risk. Training and deployment use the same exact hard matrix, so nothing is approximated at inference.

−86 bp

lower volatility than Ledoit–Wolf
significant at 99% (HAC Diebold–Mariano)

14.04% → 13.76%

annualized volatility, top-100 stocks
learned vs published threshold

0.87 → 0.76

portfolio turnover
Sharpe 0.44 → 0.54

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Let the portfolio's own risk choose the threshold

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

In plain terms

The problem Risk models need to know how stocks move together. The Gerber statistic counts the days on which two stocks move "meaningfully" in the same or opposite direction, and "meaningfully" has always been a threshold set by hand (0.50).

The idea Learn that threshold from the realized risk of the portfolio it produces. The count only changes when a stock crosses the threshold, so it has no usable gradient; DiffGerber borrows a smooth gradient for training only, while the matrix it trains is exactly the one deployed.

The result On the 100 largest US stocks the learned threshold cut minimum-variance portfolio volatility from 14.04% to 13.76% and turnover from 0.87 to 0.76, beating standard estimators such as Ledoit–Wolf (–86 bp).

MY ROLE · WHERE THIS COULD BE USED

My role First author: led the method, the experiments and the writing.

Use case Any risk or allocation pipeline that estimates a covariance matrix; a drop-in change that also trades less.

Terms used in this deck

COVARIANCE MATRIX

How much each pair of stocks moves together; the core input of risk models and minimum-variance portfolios.

GERBER STATISTIC

A robust version of correlation that counts only moves larger than a threshold, ignoring small noise.

THRESHOLD c

How big a daily move must be, in units of the stock's own volatility, to count. The paper learns it instead of fixing it at 0.50.

LEDOIT–WOLF

The industry-standard shrinkage estimator, used here as the main benchmark.

TURNOVER

How much of the portfolio is traded at each rebalance; lower means lower trading cost.

bp

Basis point: 0.01 percentage point.

A threshold you cannot backpropagate through

The Gerber statistic is a robust co-movement measure, but its key parameter is set by hand.

Why it matters

- Each day an asset is labeled up, down or neutral by whether it moves more than c times its own volatility.
- The threshold c decides signal vs noise. The published default is 0.50, and results are sensitive to it.
- Labels flip only when c crosses an observed return, so the output is a step function with zero gradient almost everywhere.
- Workarounds fall short: grids are coarse, finite differences scale with parameter count, and smooth relaxations train a matrix that is not the one deployed.

WHAT THIS MEANS OUTSIDE THE PAPER

Hard where it counts the estimator stays exact; only the gradient is relaxed.

Trained on the decision the loss is realized portfolio risk, not matrix fit.

Generalizes to any pipeline with a threshold or a discrete step in the middle.

Data & setup

RETURNS

Daily total returns and point-in-time membership, 2015–2026

EPISODES

114 non-overlapping monthly episodes, 2017–2026; 252-day estimation, 21-day outcome

UNIVERSES

Sector-balanced $N = 30 / 70$, the same plus a 9-ETF sleeve, market-cap top-100, held-out mid-caps

REFIT

Annually, using only episodes whose outcome ended before that year

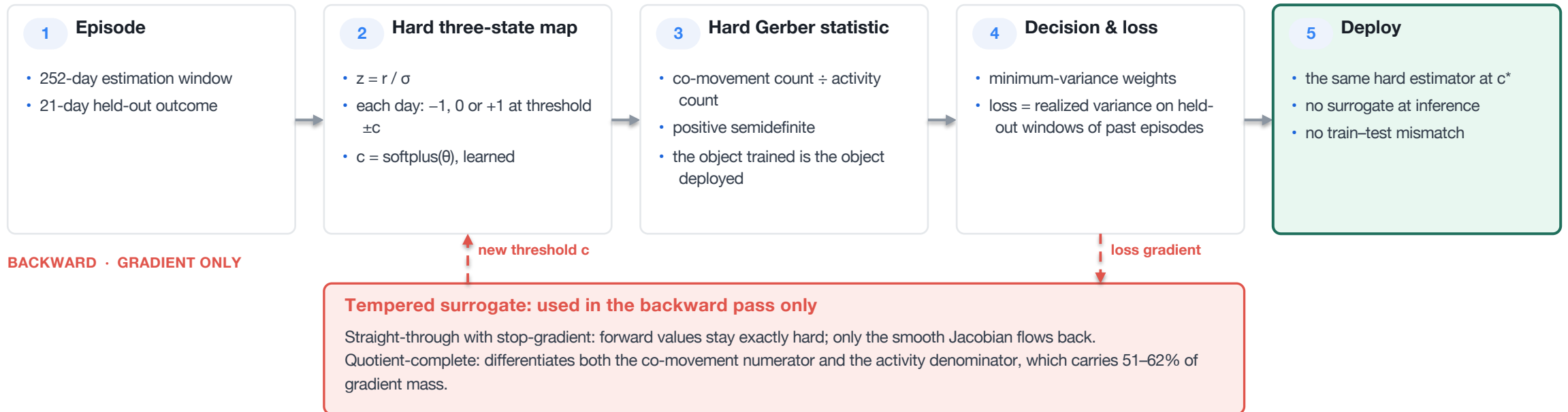
COSTS

10 bp one-way turnover cost

Hard forward, soft backward

Every forward pass uses the published hard Gerber statistic; a tempered surrogate is used only to get the gradient for the threshold.

FORWARD · EXACT HARD STATISTIC, EVERY EPISODE



Variants

DG-GMV (scalar c), DG-Asym (up/down thresholds), DG-Shrink (plus a shrinkage blend), DG-HRP, DG-Group (per asset class).

No look-ahead

Annual refit; outcome returns enter only the training loss of past episodes.

Cost

One forward and one reverse pass per update; finite differences would need 706 evaluations for a 353-parameter model.

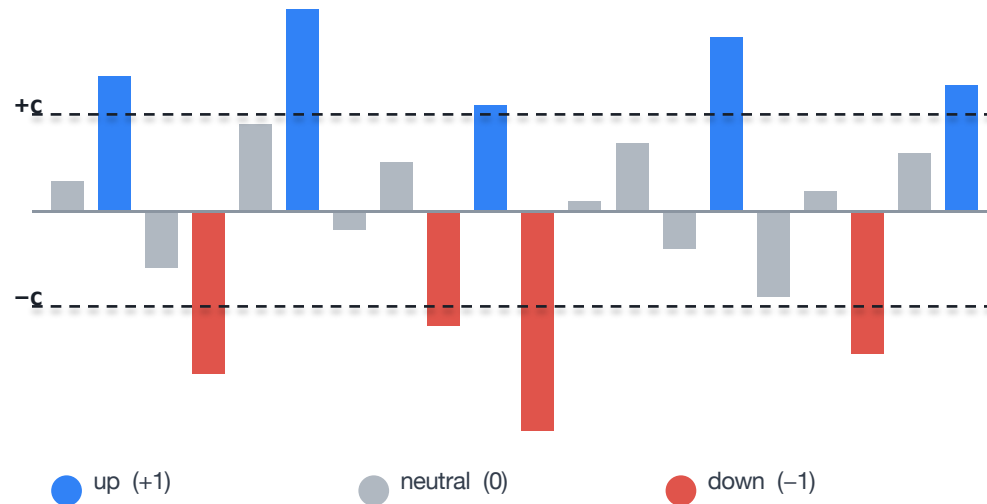
Redrawn and simplified from the paper's Figure 1.

What the threshold does, and why it blocks gradients

The threshold c turns each daily move into up, down or neutral; nudging c changes nothing until a label flips.

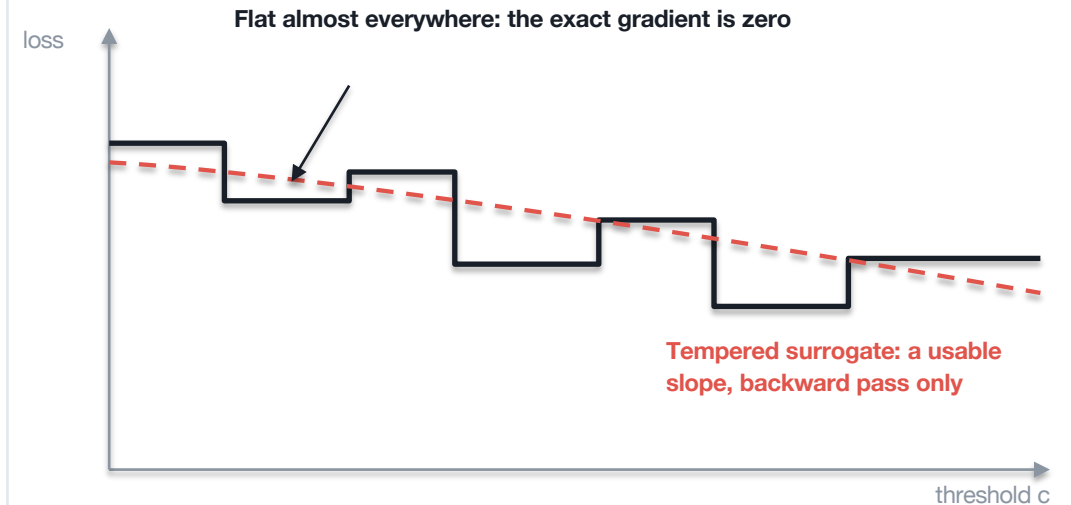
The threshold labels each day

Standardized daily moves z ; the band between $-c$ and $+c$ counts as noise



So the loss is flat in c

Portfolio loss as the threshold moves



Forward: the exact hard statistic

Backward: the surrogate's gradient

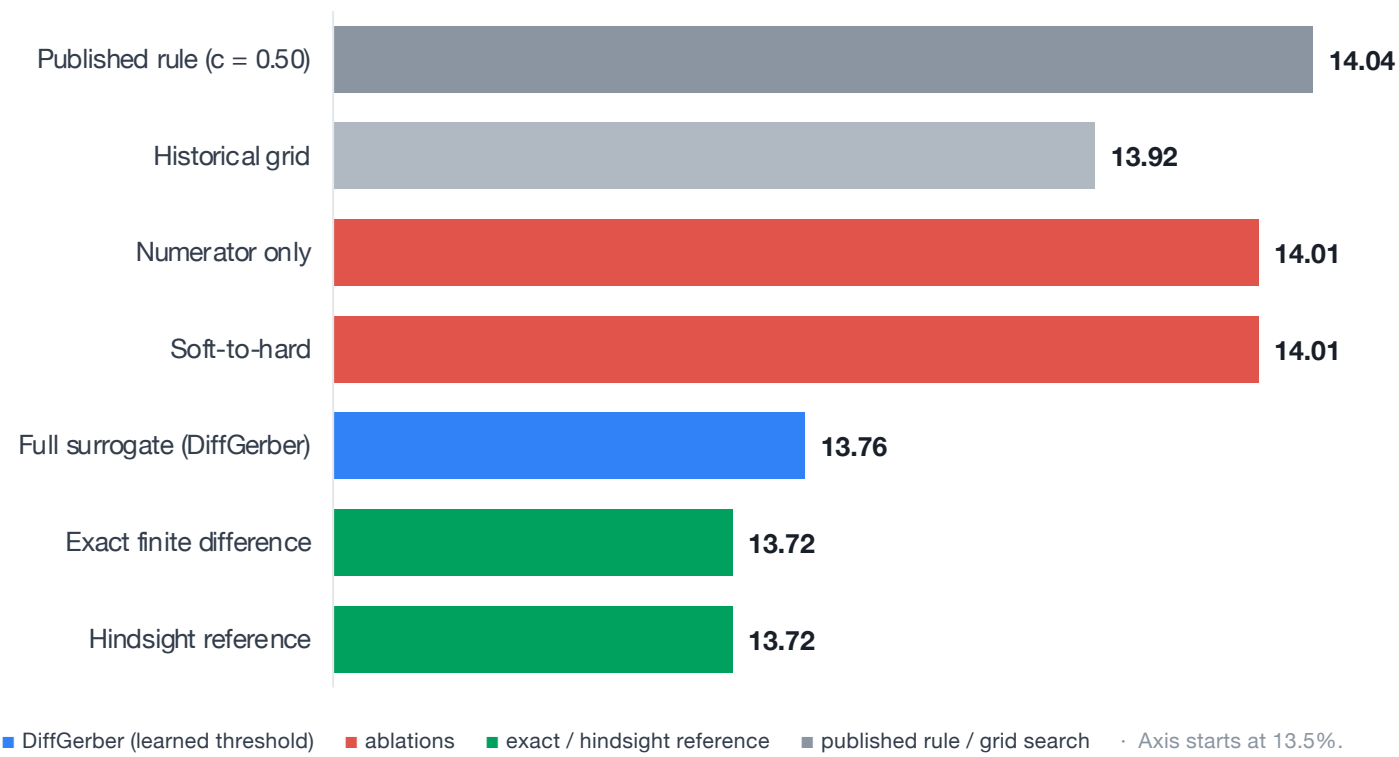
Deploy: the same hard matrix

Schematic illustration; not data from the paper.

The approximate gradient lands where exact search does

Top-100 universe, annualized volatility (%) by how the threshold is learned. Lower is better.

Annualized volatility (%)



Source: DiffGerber Table 3 and §4.4–4.5.

What it shows

- An independent exact-objective optimizer reaches 13.72%, the same narrow threshold region (learned c 1.09 vs 0.99).
- Dropping the denominator term, or training on soft states, degrades volatility to 14.01%.
- The denominator carries 51–62% of total gradient magnitude at the warm end of the temperature schedule.

Lower risk and lower turnover than standard estimators

Market-cap top-100, 114 monthly episodes, 10 bp one-way costs.

Estimator	Volatility ↓	Sharpe ↑	Max drawdown ↓	Turnover ↓
DG-GMV (DiffGerber)	13.76%	0.54	-28.8%	0.76
DG-Shrink	13.70%	0.51	-27.1%	0.82
Fixed Gerber (c = 0.50)	14.04%	0.44	-29.5%	0.87
Analytical nonlinear shrinkage	14.27%	0.51	-28.2%	1.24
Ledoit-Wolf	14.62%	0.43	-29.9%	1.54
Hierarchical risk parity	15.54%	0.62	-31.7%	0.32
Equal weight	17.78%	0.66	-33.3%	n.a.

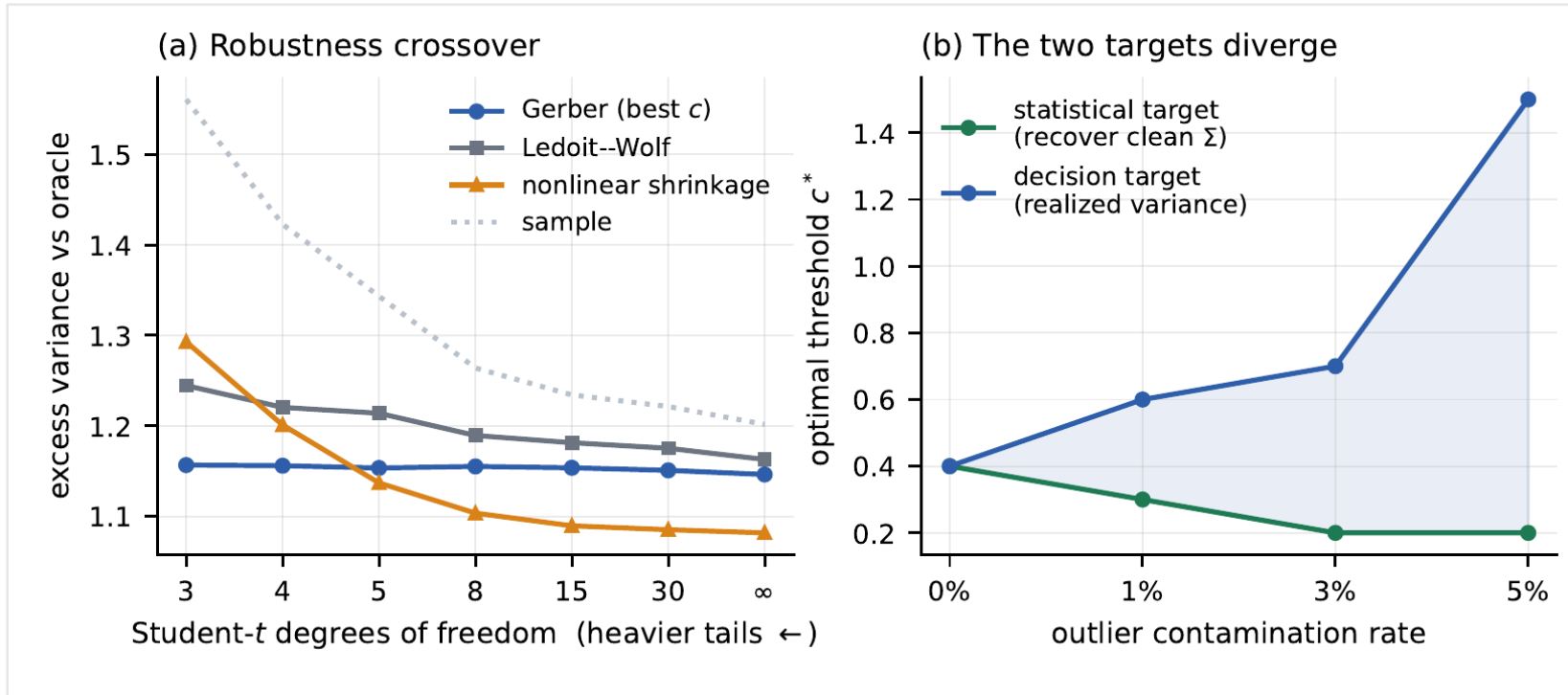
What it shows

- Volatility is 86 bp lower than Ledoit-Wolf, significant at 99% (HAC Diebold-Mariano).
- Mixed equity + ETF universes: DG-Group 4.31% → 3.92% and 4.46% → 4.00%, both significant at 1%.
- Caveat: in equity-only universes, the gain over the fixed rule is directional, not statistically distinguishable.

Source: DiffGerber Tables 1–2 and §4.3–4.4.

Why learn from the decision, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



Paper Figure 2: Controlled robustness and target separation

Left: under heavier tails it stays near the oracle while the others degrade. Right: the two targets move apart as contamination rises.

Where it helps, and where it does not

The paper reports three negative controls and a held-out null alongside the gains.

Checks that held

- Training on the portfolio decision moves c to 0.89–1.23 (13.76% volatility); training on matrix fit keeps it at 0.46–0.51 (14.08%). The decision target is what matters.
- The learned threshold tracks mean pairwise correlation (0.96 from 2019 on).
- Every learned variant beats equal weight on realized variance at the 1% level.
- Annual refit on past episodes only; code and result files are released.
- In the equity-only design, DG-Shrink reaches the lowest volatility at both universe sizes.

Negative controls and limits

- Sharpe as the training loss drives c to 0.012; out-of-sample Sharpe 0.30 vs 0.54.
- Long-only decisions: learning makes volatility worse (15.00% vs 14.84%).
- A 353-parameter state network overfits ([14.04, 14.44] vs [13.73, 13.87] for the scalar learner).
- Held-out mid-caps: learned c 0.49, no significant gain. For one scalar, finite differences are faster (5.2 s vs 9.4 s).

Code (anonymous review repository): anonymous.4open.science/r/DiffGerber-ED0B